MIL-A-36059 (BuMed)
31 May 1961

MILITARY SPECIFICATION

APRON, X-RAY PROTECTIVE

This limited coordination military specification has been prepared by the Bureau of Medicine and Surgery (MMSA) based upon currently available technical information. It is subject to modification. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers one type of lead-impregnated plastic apron, suitable for use as X-ray protection during radiological procedures.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

CCC-T-191	Textile Test Methods.
PPP-B-585	Boxes; Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.

Military

MIL-T-7807	Thread, Nylon.
MIL-L-10547	Liners, Case, Waterproof.

STANDARDS

Federal

Fed. Test Method Std. No. 601	Rubber: Sampling and Testing.
Fed. Std. No. 751	Stitches, Seams, and Stitchings.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Military Medical Supply Agency

20400 Apron, X-Ray Protective.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Material.

3.1.1 Plastic sheet. Plastic sheet shall be single- or multiple-ply flexible vinyl, uniformly impregnated with lead or lead compound to provide a lead equivalent of not less than 0.8 millimeter (mm.), when tested as specified in 4.4.1. The impregnated plastic sheet may be faced front and back with vinyl plastic bonded thereto. The finished sheet shall have the following properties, when tested as specified in 4.4.1.

Weight per square foot, pounds, maximum	2.75
Tensile strength, minimum	
Original, pounds per square inch (p.s.i.)	600
After aging, percent of original	75

The apron body shall show no physical change or loss of radiation protective value, when tested as specified in 4.4.1.1.2. Color shall be white or cream. Outer surfaces shall be smooth and washable.

3.1.2 Cloth. Cloth used for the shoulder pieces shall be cotton and shall have the following characteristics, when tested as specified in 4.4.2.

Weight, ounces per square yard, minimum	2.70
Thread count per inch, minimum	
Warp	66
Filling	45
Breaking strength, pounds, warp and fill, minimum	40
Color	Brown

3.1.3 Tape. Tape shall be cotton twill weave, 1 inch +1/16,-1/32 inch, wide, having the following characteristics, when tested as specified in 4.4.3.

Thread count	
Ends, minimum	80
Filling, per inch, minimum	38
Breaking strength, pounds, minimum	100
Color	Brown

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3.1.4 Thread. The thread shall be nylon, conforming to type I, class 1, size E of Specification MIL-T-7807. The color shall be brown to match the color of the tape.

3.2 Design and construction.

3.2.1 Apron body. Apron body shall be made from one piece of plastic sheet. When laid out flat, the apron body shall not be less than 24 inches wide and 38 inches long, measured from the low point of the neck to the bottom of the apron body. The sides of the apron body shall extend not less than 5 inches above the bottom of the neck cutout. The radius of the neck cutout shall be 5 inches ± 1/2 inch. All corners of the apron body shall be rounded. The edges of the apron body shall be bound to not less than 6 inches from the bottom of the shoulder piece, in accordance with Drawing 20400.

3.2.2 Shoulder pieces. The shoulder pieces shall be of such design that when attached to the apron body, a form-fitting coat, properly cut out for arms and shoulders, will result and the weight of the apron body will be distributed throughout the shoulder pieces. The shoulder pieces shall be attached to the apron body with double stitching. All edges of the shoulder pieces shall be bound with tape (see 3.1.3).

3.2.3 Ties. There shall be two sets of ties constructed of tape as specified in 3.1.3 and connected on the rear of the shoulder pieces. One set, each tie a minimum of 20 inches long, shall be securely stitched midway between the waist and the top of the shoulders. A second set, each tie a minimum of 40 inches long, shall be securely stitched at the waist line. The free ends of the tie tapes shall be folded back 1/2 inch. The raw edge shall be turned under 1/4 inch, then stitched 1/8 inch from the edge, using stitching type EFb-1 conforming to Federal Standard No. 751.

3.3 Stitches, seams, and stitchings. Stitches, seams, and stitchings shall conform to Federal Standard No. 751. Type 301 stitches with five to seven stitches per inch shall be used. Binding shall be attached using seam type BSa-1. Shoulder pieces and tapes shall be attached to the apron body with two rows of stitching 3/16 inch apart. The first row shall be on the binding 1/8 inch ± 1/16 inch from the selvage edge (see drawing).

3.4 Identification marking. The inside surface of the apron body shall be permanently and legibly marked with the name or registered trademark of the manufacturer, lead equivalent of the apron, and the X-ray tube peak voltage to which the lead equivalent applies.

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3.5 Workmanship. Workmanship shall be first class throughout. If facing is used on the apron body, the adherence shall be such as to give the appearance of one-ply construction. The binding shall be free of raw edges. Apron shall be free from defects which detract from its appearance or which may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling. Sampling shall be conducted in accordance with Standard MIL-STD-105.

4.2.1 For examination. The lot size shall be expressed in terms of one apron and the sample unit shall be one completely fabricated apron.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	II	2.5
Total (major and minor defects combined)	II	6.5
For dimensional examination	L6	4.0
For examination of preparation for delivery requirements	L4	4.0

4.2.2 For tests. Sampling for tests shall be in accordance with table I.

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Table I - Component Testing

Component and lot size expressed in terms of	Characteristic	Rqmt para	Test para.	Rqmt. Appl to: Indiv Unit	Rqmt. Appl to: Lot Avg	No. of det per smpl unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Smpl unit	Insp level	AQL
Plastic sheet (1 yd)	Lead distribution	3.1.1	4.4.1.1.1	X	-	1	-	0.01 min	Two 1 ft sq pieces	(*)	(*)
	Radiation deterioration	3.1.1	4.4.1.1.2	X	-	1	X	-	"	(*)	(*)
	Tensile strength:										
	Original	3.1.1	4.4.1.1.3	X	-	1	-	1.0 p.s.i	½ yd	L2	4.0
	After aging	3.1.1	4.4.1.1.3	X	-	1			"	"	"
	Weight per sq ft	3.1.1	4.4.1.1.4	X	-	1	-	0.01 lb	"	"	"
Cloth (for shoulder pieces) 1 yd full width	Weight oz/sq yd	3.1.2	4.4.2	X	-	5	-	0.1 oz	1 yd	L1	6.5
	Thread count:										
	Warp	3.1.2	4.4.2	X	-	5	-	Whole No.	"	"	"
	Filling	3.1.2	4.4.2	X	-	5	-	Whole No.	"	"	"
	Breaking strength:										
	Warp	3.1.2	4.4.2	X	-	5	-	1.0 lb	"	"	"
	Filling	3.1.2	4.4.2	X	-	5	-	1.0 lb	"	"	"
	Color	3.1.2	Visual	X	-	1	X	-	"	"	"
Tape (1 linear yd)	Thread count:										
	Ends	3.1.3	4.4.3	X	-	1	-	Whole No.	1 yd	L1	6.5
	Filling	3.1.3	4.4.3	X	-	1	-	Whole No.	"	"	"
	Breaking strength	3.1.3	4.4.3	X	-	5	-	1.0 lb	"	"	"
	Color	3.1.3	Visual	X	-	1	X	-	"	"	"

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(*) The contractor shall test each roll of plastic sheet for compliance with this characteristic. For Government verification testing the inspection level shall be I4. Any failure encountered in testing will be cause for rejection of the lot represented by the sample.

4.3 Examination. The apron shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classifications of defects.

NOTE: When asterisk appears in the Major defects column, the defect should be classified as follows:

Major: When seriously affecting serviceability.
Minor: When affecting serviceability but not seriously.

EXAMINE	DEFECT	MAJOR	MINOR
Color (applicable to all components)	Not specified or approved color		X
Cleanliness	Grease or oil stains or other form of foreign matter which cannot be readily removed	X	
	Not clean; i.e., dirty, but can be cleaned readily with a cleaning agent		X
Design	Any characteristic not in accordance with specified requirements (unless otherwise indicated herein)	X	
Material (general)	Not fabricated of specified material	X	
(Apron body)	Not flexible	X	
Construction and workmanship (applicable to all components and assemblies, unless otherwise indicated herein)	Any cut, hole, or rupture	X	
	Any blister, burn, pinch, pit, thin or hard spot, abraded area, crease, or similar defective condition	(*)	
	Any repair or patch	(*)	

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EXAMINE	DEFECT	MAJOR	MINOR
Construction and workmanship (Cont)	Any solid plastic ridge, run, or lump		X
	Evidence of delamination (if applicable)	X	
	Tacky	X	
	Embedded foreign matter which when removed will reduce thickness of apron body	(*)	
	Foreign inclusion which cannot be readily removed		X
	Not constructed as specified or in accordance with approved sample	X	
	Corners not rounded where required		X
	Any component missing	X	
	Any operation omitted or not properly performed	(*)	
	Any component or part misaligned, malformed, or damaged	(*)	
Seams and stitching	Not specified seam type	X	
	Not specified stitch type	(*)	
	More than maximum number of stitches per inch damaging material	X	
	More than maximum number of stitches per inch but not damaging material		X
	One stitch per inch less than the specified minimum		X
	Two or more stitches per inch less than the specified minimum	X	
	Any stitching omitted where required	(*)	
	Loose stitch tension resulting in a loosely secured seam	(*)	
	Tight stitch tension	(*)	

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EXAMINE	DEFECT	MAJOR	MINOR
Seams and stitching (cont.)	Open seam		
	NOTE: When one or more stitches are broken or when two or more continuous skipped stitches or runoffs occur on a single stitched seam or on both rows of stitching in a double stitched seam, it shall be classified as a major defect. When one or more stitches are broken or when two or more continuous skipped stitches occur on only one row of stitching in a double stitched seam, it shall be classified as a minor defect.		
Marking	Missing, incorrect, incomplete, illegible, not in approved location, not approved color, or not accomplished as specified		X

4.3.2 Dimensional examination. The aprons shall be examined for defects in dimensions. Any dimension that is not within the established tolerance shall be classified as a defect. For those dimensions where no tolerance is indicated, the following shall apply:

For dimensions up to 5 inches - ± 1/16 inch
For dimensions 5 to 15 inches - ± 1/2 inch

4.3.3 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing, and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one exterior container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed below shall be examined on exterior containers fully prepared for delivery. The lot size shall be the number of exterior containers in the end item inspection lot. The inspection level shall be I4 and the AQL shall be 4.0 defects per 100 units.

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Examination	Defect
Markings (exterior and interior)	Incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component; component missing, damaged, or otherwise defective affecting serviceability
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, etc.
	Bulging or distortion of containers.
Weight or content	Number of intermediate packages is more or less than required: gross net weight exceeds requirement.

4.4 Tests. Tests shall be conducted, as necessary, to determine compliance with specification requirements.

4.4.1 Plastic sheeting.

4.4.1.1 Radiographic tests. Radiographic tests to determine compliance shall be carried out using 88 kilovolts-peak X-rays. The following technical factors are suggested for use in carrying out the tests:

88 kilovolts-peak.
10 milliamperes.
60 seconds.
30 inches distance.
Standard X-ray films.

4.4.1.1.1 Distribution. The entire surface of the leaded vinyl plastic apron shall be placed alongside a lead penetrometer having 10 steps per mm. of thickness. The minimum thickness of the penetrometer shall be 0.1 mm. or less, and the maximum thickness 0.8 mm. or more. After development, a comparison of densities shall be made by means of a suitable densitometer.

4.4.1.1.2 Radiation deterioration test. Lead-impregnated vinyl plastic sheet shall be subjected to massive X-ray radiation in the range of 200,000 roentgens.

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4.4.1.1.3 Tensile strength. Tests for tensile strength shall be conducted in accordance with method 4111 of Federal Test Method Standard No. 601.

4.4.1.1.4 Weight. The weight shall be determined by weighing a one-square-foot sample, accurate to the nearest 0.01 pound.

4.4.2 Cloth. Tests for cloth shall be conducted in accordance with the following test methods described in Specification CCC-T-191:

Characteristic	Method
Weight	5041
Thread count	5050
Breaking strength	5100

4.4.3 Tape. Tests for tape shall be conducted in accordance with the following test methods described in Specification CCC-T-191:

Characteristic	Method
Thread count	5050
Breaking strength	5100

5 PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) apron, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II - Packaging quantities

Unit package	Intermediate package	Exterior container
1 unit	Not required	6 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

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5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with Specification PPP-B-636. Each carton shall be adequately secured to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose within the continental limits ot the United States of America.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designated for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; or PPP-B-621, table III.

5.2.1.2 Case liner. Each level A box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

5.2.1.3 Strapping. Strapping, when required, shall be in accordance with applicable box specification.

NOTE: Strapping will not be required for shipment forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to Specification PPP-B-601, table I, using type III, class 1 plywood, or shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type I, class 1, fabricated from not less than 275-pound test fiberboard.

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5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or order.

5.3 Marking.

5.3.1 Unit packages. Each unit package shall be marked in accordance with 5.2 of Standard MIL-STD-129. In addition, contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Military requirements.

6.3 Transportation description. Transportation description applicable to this item is:

Plastic articles, noibn, not cellular, expanded, nor foamed, weighing 12 pounds per cubic foot or over, in boxes.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army-Med.
Navy-BuMed
Air Force-APCSG

Preparing activity:
Navy-BuMed
(MMSA)

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